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Data Driven Production Choke Flow Correlation: Application in Virtual Flow Metering and Well Test Optimization

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Abstract

This paper aims to develop a data-driven enhancement of traditional choke flow correlations using a swarm-intelligence-based optimization algorithm. The goal is to accurately predict daily production rates of naturally flowing oil wells and streamline the well-test selection process, offering a scalable and cost-effective alternative to flowmeter deployment and complex simulation based virtual flow metering systems.

The approach integrates the empirical choke flow model by Gilbert et al. with historical well-test data to establish a dynamic, data-driven relationship between surface operating parameters and oil flow rates. A Python-based stochastic optimization algorithm is employed to continuously adjust the correlation coefficients, enabling real-time adaptability to changing production conditions. The optimized model is then applied in a virtual flow metering workflow to estimate daily production rates and identify high-priority candidates for well testing. Field data from naturally flowing wells are used to train, validate, and benchmark the model's performance against traditional empirical methods and simulation-based estimators.

The optimized choke flow correlation significantly outperforms traditional fixed-coefficient models, delivering improved prediction accuracy under varying operational conditions. Field implementation demonstrates that daily flow rate estimates generated using this method closely match actual test data, even wells with limited or infrequent historical tests. Additionally, the model enables automated identification of wells most in need of testing, improving the efficiency of surveillance operations and reducing the likelihood of missed diagnostic opportunities. By minimizing the reliance on manual scheduling and high-capex simulation systems, this method supports more timely and cost-effective reservoir management. The results confirm that a data optimized empirical model offers a powerful alternative for virtual metering and well-test optimization in resource-constrained environments.

This paper introduces a novel hybrid methodology that merges empirical models with AI-driven optimization, providing a practical, low-cost solution to enhance production surveillance. It advances the state of knowledge by improving the adaptability of choke correlations for real-time applications and bridging the gap between data science and field operations.

Introduction

Reliable estimation of well-level production rates is essential for production surveillance, reservoir management, and well-test planning. However, direct measurements using multiphase flowmeters are limited in field-wide deployment due to their high capital cost, maintenance complexity, and long-term reliability issues [1]. As a result, operators often rely on allocation techniques based on intermittent well tests and operating parameters, which introduce significant uncertainties due to infrequent testing and variable flow behavior [2].

Virtual Flow Metering (VFM) has emerged as a scalable solution, offering production rate estimation without physical flowmeters. While simulation-based VFMs are powerful, they are computationally intensive and dependent on accurate boundary conditions and calibrated models [3]. In contrast, empirical correlations, particularly those derived from choke flow behavior remain widely used for their simplicity and speed. Among them, the correlation proposed by Gilbert (1954) [4] has endured as the industry benchmark for estimating liquid flow rates under critical flow conditions at the wellhead.

A general form of the equation can be defined as shown in Equation 1.

$$q_l = \left(\frac{1}{10} \times \frac{P_{wh} D^{1.89}}{R^{0.546}} \right)$$

Equation 1—Gilbert’s empirical choke flow correlation`

Where:

q_l : Liquid flow rate, bbl/d

P_{wh} : Wellhead pressure, psia

D : Choke size, 1/64 inches

R : Gas-Liquid Ratio (scf/bbl)

Over the years, several researchers have proposed alternative correlations using the same governing variables but with different empirical coefficients to match their field-specific datasets. The generalized form of such correlations is shown in Equation 2.

$$q_l = \left(\frac{1}{A} \times \frac{P_{wh} D^C}{R^B} \right)$$

Equation 2—General form of empirical choke flow correlation

Table 1—Values of coefficients for empirical choke flow correlation [5]

Empirical Choke Flow Correlation	Year	A	B	C
Gilbert Equation	1954	10	0.546	1.89
Baxendell Equation	1957	9.56	0.546	1.93
Ros Equation	1960	17.4	0.5	2
Achong Equation	1961	3.82	0.65	1.88

These models, while simple, lack adaptability to changing field conditions because of their static coefficients. As a result, their accuracy declines over time, especially in wells with variable GOR, pressure, or flow regime. However, with the increasing availability of historical production and test data, data science offers an opportunity to dynamically tune these coefficients to better reflect real-time conditions [5].

This paper proposes a novel enhancement to Gilbert’s correlation using a swarm intelligence-based optimization algorithm that learns from field data to calibrate coefficients continuously. This dynamic, well-specific approach significantly improves daily flow rate estimation. When the approach cannot estimate the

accurate rates due to very old well test or invalid well test data, it tags those wells as high-priority candidates for production testing [6].

Methodology

Workflow Design and Rationale

This study adopts a well-specific, data-driven approach to optimize existing choke flow correlations rather than derive new ones. Traditional empirical models like Gilbert's have served the industry for decades, but their static nature makes them less effective under changing flow regimes. The proposed method focuses on tuning the existing empirical coefficients based on recent well-test data, allowing the model to respond to evolving well conditions without the need for simulation or extensive instrumentation.

Unlike most generic approaches, the developed workflow does not aim for a universal solution. Instead, it calibrates a tailored model for each well using its own historical data. This not only accounts for well-specific characteristics but also adapts to dynamic reservoir behavior around the well. The result is a set of optimized coefficients (A, B, and C) that are unique for each well and can be updated as new well-test data becomes available. The entire process is orchestrated through Python-based scripting, enabling seamless integration with cloud-based and on-premises production databases and dashboards.

A schematic representation of the workflow shown in Fig. 1. includes four major components: data gathering, preprocessing, coefficient optimization using Particle Swarm Optimization (PSO), and rate prediction. The model requires a minimum of five recent well-test records per well for training, along with at least one valid set of operating conditions for prediction. This user-defined threshold ensures flexibility and adaptability across assets with varying data availability. Wells where the algorithm fails to converge, could indicate towards invalid well test data and are highlighted for further study (potential well test candidates).

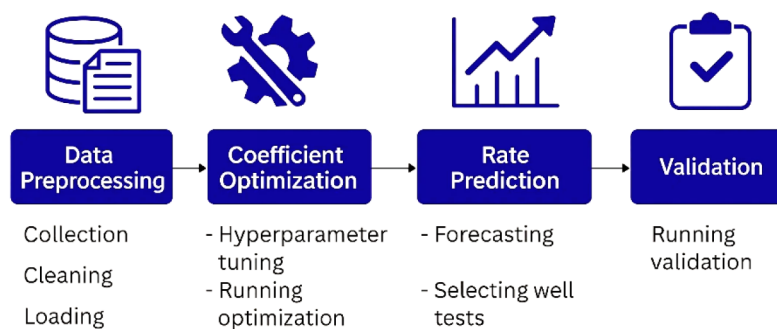


Figure 1—Virtual flow metering workflow

Particle Swarm Optimization (PSO)

PSO is a population-based optimization technique inspired by the social behavior of animals that move in groups, such as birds flocking or fish schooling. It is a type of evolutionary computation method that iteratively searches for the best solution in a multidimensional space. Unlike gradient-based optimization methods, PSO does not require the function being optimized to be continuous or differentiable. This makes it particularly useful for a wide range of real-world problems where the objective function is noisy, discontinuous, or non-analytical.

In PSO, each particle in the swarm represents a possible solution to the optimization problem. A particle is essentially a point in the solution space, and the goal is to adjust its position iteratively to move closer to the global optimum. Each particle maintains three key attributes throughout the optimization process:

1. Current position in the solution space p_i
2. Velocity to determine its direction and speed of movement v_i

3. The memory of the best position it has found so far, known as its personal best or $pbest_i$

Additionally, every particle has access to the best position discovered by any particle in the swarm, referred to as the global best or $gbest_i$. The visual representation provided helps capture this idea is shown in Fig. 2.

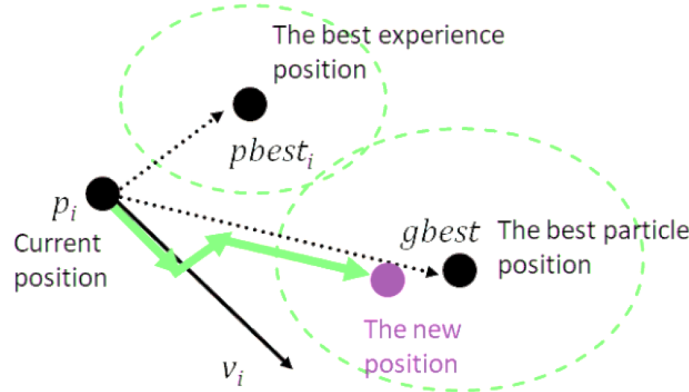


Figure 2—PSO working principle

The Fig. 2 shows a particle at its current position p_i . From there, it is influenced by two key attractors:

1. Its own historical best $pbest_i$, which represents the best solution it has found so far.
2. The global best $gbest_i$, which is the best solution any particle in the entire swarm has identified up to that point.

The movement of the particle is determined by a velocity vector v_i , which is updated each iteration based on a weighted combination of three components:

1. Its inertia (previous velocity)
2. The cognitive component (toward $pbest_i$)
3. The social component (toward $gbest_i$).

This results in a new position, indicated by the purple dot in Fig. 2.

The behavior of the particle is governed by two guiding principles: self-exploration and cooperation. Self-exploration is the process by which each particle tries to find better positions independently, guided by its own best experiences. Cooperation is achieved when particles adjust their movement based on the experience of the swarm as a whole. The interaction between these principles helps the particles avoid getting stuck in local optima and instead move toward the global optimum.

The velocity update equation captures this dynamic interaction and is usually defined as:

$$v_f(t+1) = w \cdot v_f(t) + c_1 \cdot r_1(pbest_i - x_f(t)) + c_2 \cdot r_2(gbest_i - x_f(t))$$

Here, w is the inertia weight that controls how much of the previous velocity is retained, c_1 and c_2 are acceleration constants for the cognitive and social components respectively, and r_1 and r_2 are random numbers between 0 and 1 to ensure stochastic behavior. The position of each particle is updated using:

$$x_f(t+1) = x_f(t) + v_f(t+1)$$

This balance between exploration and exploitation is key to PSO's efficiency. A higher inertia weight encourages global exploration, while a lower one facilitates local exploitation. Typically, the inertia weight is gradually reduced over time, allowing the swarm to explore broadly during early iterations and then converge more precisely during later ones.

One of PSO's greatest strengths lies in its simplicity. It does not require complex mathematical operations or problem-specific knowledge, and its computational cost is relatively low. This has made PSO popular in fields ranging from neural network training and image processing to structural optimization.

In the context of production systems or flow estimation, PSO can be particularly useful when optimizing models that relate well performance to choke size, gas-liquid ratio, and pressure parameters. As these relationships are often non-linear and affected by time-varying flow regimes, traditional empirical models like Gilbert's correlation may not yield accurate results across all conditions. By applying PSO, one can dynamically adjust the equation's coefficients based on recent field data, thereby improving accuracy and making the model more responsive to changing operational conditions.

To illustrate, imagine a well operating under different flow regimes over time due to reservoir pressure depletion or water breakthrough. A static choke correlation using fixed coefficients may consistently over- or under-estimate flow rates. PSO, on the other hand, can be used to optimize those coefficients based on the latest well test data. The algorithm tunes the values in a way that minimizes the difference between predicted and actual production rates. Over time, as new data becomes available, the coefficients are re-optimized, and the model evolves with the well's behavior, ensuring consistent performance and reliability.

Data Requirements and Processing

For successful implementation, the model requires both training and prediction datasets. Training data includes valid well-test entries with measured liquid rate, gas-liquid ratio, choke size, and pressure parameters. Prediction data involves the same operational inputs but excludes the measured rate, which the model aims to estimate.

To validate the workflow, two datasets were used:

1. **Simulated Data:** Created using PIPESIM, this synthetic dataset models a naturally flowing horizontal oil well under various flow conditions by performing sensitivity runs across multiple operational parameters. It serves as a controlled environment to test model behavior against known outputs.
2. **Field Data:** Sourced from the North Sea Transition Authority's public Banff oilfield dataset, this real-world data includes over 1,000 well-day records. Production volumes, pressures, and choke settings were cleaned, transformed, and derived into model-ready inputs. To ensure reliability, only data from the most recent five-year window (2015–2019) was used [7] [8].

The preprocessing pipeline involves null-value removal, unit conversion, and derivation of essential inputs such as choke size (in 1/64 inches), gas-liquid ratio, and wellhead pressure (in psia). These are standardized using a set of transformation equations and thresholds designed to ensure consistency across both datasets.

Optimization coefficients (A, B, and C) using PSO

Once the data is prepared, the core objective is to optimize the empirical coefficients of the choke flow model PSO. The optimization of coefficients value (A, B, and C) is executed for one well at a time using its respective well-test history. Each test record provides a set of input conditions and a measured flow rate, which are used to minimize the error between predicted and actual rates by adjusting the coefficients.

A significant challenge is the limited amount of valid data per well. Well tests are often infrequent, and older tests may not reflect current well conditions. As such, only the most recent 5–10 well tests are recommended for optimization, keeping the model focused on relevant operational states.

To tackle this data constraint, PSO is employed for its effectiveness in low-data, nonlinear optimization problems. The algorithm uses a swarm of candidate solutions (particles) that iteratively explore the search space, learning from both individual and group performance to find the best set of coefficients. The process continues until the prediction error across all training points is minimized below a defined threshold.

Enhancements to Standard PSO

Two major modifications were introduced to the standard PSO framework to improve performance and applicability for production surveillance tasks:

1. **Temporal Weighting:** Traditional PSO assigns equal weight to all training data. However, recent well tests are more reflective of current reservoir and surface conditions. A weighted cost function was developed, giving exponentially higher importance to recent data. This allows the model to adapt faster to changes in well performance while retaining robustness.
2. **Smart Initialization:** Instead of random initialization for all particles, a subset is seeded with known empirical coefficients from established choke flow models (e.g., Gilbert, Baxendell, Ros). This increases convergence speed and improves search efficiency by anchoring the optimization process to viable starting points known to perform reasonably well across various conditions.

These enhancements transform PSO to operate on time sequence data, which is needed for this problem statement. All optimization parameters such as population size, iteration count, learning coefficients, velocity constraints, and convergence tolerances are configurable. Default values are selected based on testing across multiple wells, but users can fine-tune these settings based on field-specific needs. Once, the new choke flow coefficients are identified after optimizing it over well test data, these coefficients will be used in Equation 2 for VFM. Additional details related to method is available in patent application by authors [9].

Results and Discussion

To evaluate the performance and reliability of the proposed approach, multiple comparative studies were conducted using both simulated and field datasets. The initial validation employed synthetic data generated from a simulation environment. Among all tested methods, the PSO-based model consistently demonstrated the lowest deviation from simulated rates, indicating strong alignment with the expected flow behavior. However, recognizing the limitations of synthetic data, which lacks the variability and noise of real-world conditions, the study also applied the workflow to a more complex dataset derived from actual field operations.

This field dataset comprised over 1000 data points collected from the field in the North Sea. Results are illustrated in Fig. 3. Out of the other empirical models evaluated, Gilbert's correlation produced the most accurate estimates, yet it was outperformed by the PSO-optimized model, which more effectively captured shifts in flow regime and operating conditions. Because the PSO framework updates its coefficients with every new valid well-test, it is inherently responsive to dynamic production behavior.

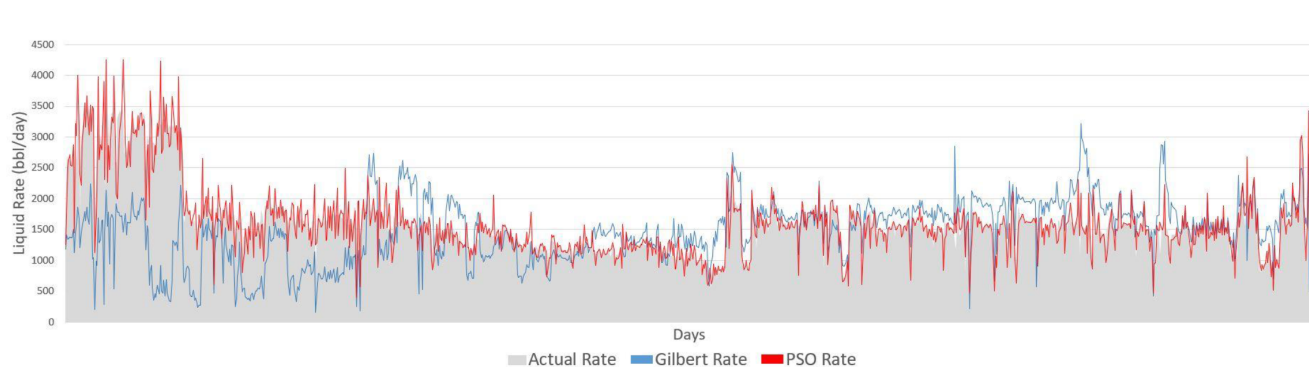


Figure 3—Comparative analysis of prediction by Gilbert's correlation and Modified PSO on the data collected from the field

The model's effectiveness is further validated by a measured vs. predicted rate comparison (Fig. 4), which reveals a strong linear correlation with a slope of 0.97. This near-unity slope suggests that the PSO-based estimates closely track actual field measurements, reinforcing the model's predictive reliability.

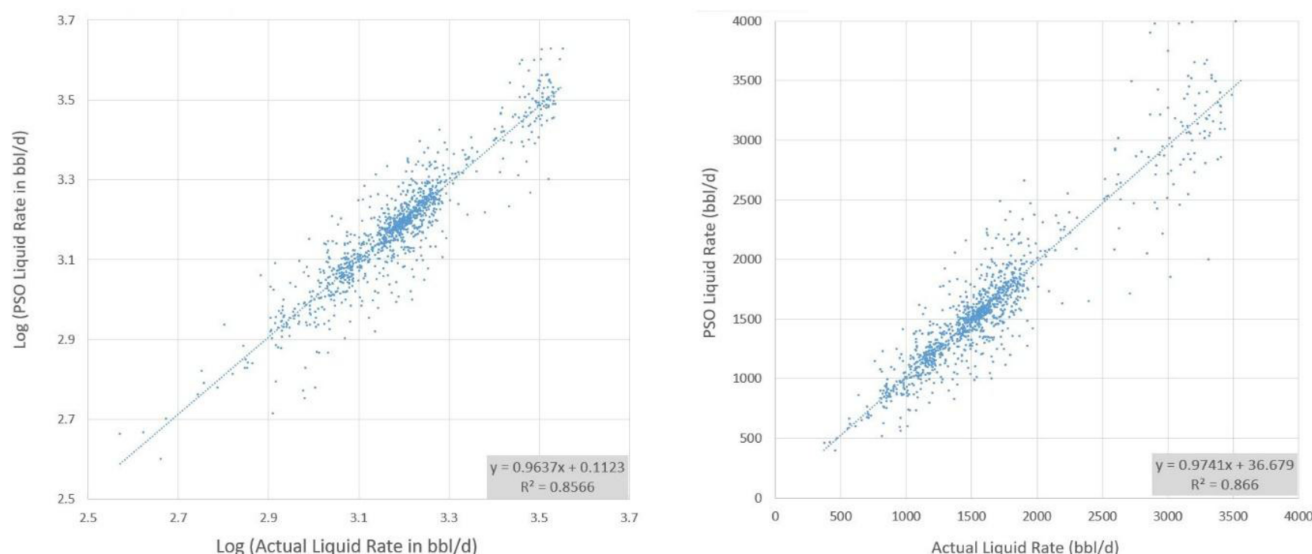


Figure 4—Actual vs PSO estimated rate comparison

To deepen the evaluation, the field dataset was segmented into three operating regimes: low, medium, and high liquid rate cases, after filtering out a small number of extreme outliers. Across all three segments, the PSO-enhanced model consistently delivered a substantial reduction in prediction error. On average, it achieved a 20-40% improvement in accuracy over the best-performing empirical correlation, further emphasizing its robustness and adaptability for use in dynamic field environments.

In addition to estimating daily production rates, the proposed PSO-based method proved valuable for identifying potential well test candidates. During optimization, if the algorithm fails to converge or produces large discrepancies between predicted and actual flow rates for a given well, it may indicate that the underlying well test data is outdated or invalid. Such wells can be flagged for re-evaluation. A ranked list of these candidates can be automatically generated based on the optimization error, with highest priority assigned to wells exhibiting the largest mismatch between modeled and observed rates. This offers a data-driven approach to prioritize well testing activities; however, final decisions should always remain with the field engineers, who can consider additional operational context. Notably, analysis of field data from the North Sea revealed that wells with persistent optimization failures were often those with very old or inconsistent well test records, further reinforcing the utility of this method as a well test validation tool in addition to flow rate prediction.

Conclusion

Estimating daily production rates remains a complex yet essential aspect of production optimization, particularly in fields with economic and operational constraints. Despite advancements in metering technology, many operators, especially those managing a large number of wells with limited simulation capacity, still rely on decades-old empirical models like the Gilbert correlation to perform VFM.

This paper presents a novel enhancement to the traditional choke flow approach by introducing a data-driven method that dynamically tunes the empirical coefficients using a modified Particle Swarm Optimization (PSO) algorithm. The proposed workflow is designed to adapt to changing flow regimes,

minimize manual calibration, and integrate seamlessly with both cloud and on-premises systems that support Python-based automation.

To validate the workflow, it was applied to both synthetic and real-world datasets, including over 1,000 field data points from the North Sea. The PSO-optimized model showed a 25–50% improvement in predictive accuracy over traditional choke correlations across varied flow conditions. Unlike static models, it adaptively refines estimates with each new well test. If PSO fails to converge or yields poor accuracy for a specific well, it may indicate outdated or unreliable test data. Such wells are flagged as potential candidates for re-testing. A ranked list can be generated, prioritizing wells with the highest prediction error, though final decisions remain at the discretion of field engineers.

In summary, the developed method offers a practical and scalable solution for improving VFM accuracy in dynamic production environments. It bridges the gap between traditional empirical models and modern data-driven techniques, enabling operators to achieve better surveillance outcomes with minimal overhead.

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